Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2022)

Course Code with Title	**CS-326: Information Security**		**Program**	Bachelor of Computer Science BS (CS)
Instructor	**Dr. Amir Mehmood**		**Semester**	6^{th}
Start date & Time	December 08, 2022 at 09:30 AM	**End Time**	11:00 AM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Questions along with their parts **should be answered to the point and in sequential order.**

Q.1. **(3+3)**

a) Suppose a user has created a social media account for data sharing and storage in which users provide personal identification, and a secret code for account access. Give examples of confidentiality, integrity, authentication and availability requirements associated with the social media system.

b) Discuss, how an attacker does traffic analysis and fabrication attack?

Q.2. **(3+3)**

a) Use Hill Cipher to Encrypt the Message

Plain Text = **RE** Key = $\begin{bmatrix} 4 & 3 \\ 2 & 7 \end{bmatrix}$

18 15

b) Use One Time Pad to Encrypt the Message

Plain text= **EXAMPLE** Key= **DOLLARS**

Q.3. **(3+3)**

a) Generate keys for S-DES algorithm from 10 bits key = 1100111101 using S-DES key generation where

P10=	3	5	2	7	4	10	1	9	8	6
P8 =	6	3	7	4	8	5	10	9		

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b) Suppose you are at second round of S-DES encryption and performing function fk. You have 8 bits 10011010 after performing swapping function. Considering Key K2 as 11010010, execute second round to obtain 8 bits cipher. where

IP =	2	6	3	1	4	8	5	7
IP^{-1}=	4	1	3	5	7	2	8	6
E/P =	4	1	2	3	2	3	4	1
P4 =	2	4	3	1				

$$S_0 = \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 \\ 3 & 1 & 3 & 2 \end{bmatrix} \qquad S_1 = \begin{array}{c} \\ 0 \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{bmatrix} 0 & 1 & 2 & 3 \\ 2 & 0 & 1 & 3 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 3 \end{bmatrix} \end{array}$$

Q.4. **(4+2)**

a) Explain the Round Function of the Data Encryption Standard (DES) with the help of a labeled diagram.

b) Let the input block B = 111011, find S6 (111011) by passing input bits from S-box 6.

Q.5. **(06)**

Find out the value of byte ($S_{2,0}$) of first column of MIX COLUMNS in the AES algorithm if output of shift rows is

$$\begin{bmatrix} 63 & 47 & A2 & F0 \\ 9C & 63 & C5 & F2 \\ 7B & 7C & F0 & AB \\ CA & AF & 76 & 76 \end{bmatrix}$$

$$\begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \begin{bmatrix} s_{0,0} & s_{0,1} & s_{0,2} & s_{0,3} \\ s_{1,0} & s_{1,1} & s_{1,2} & s_{1,3} \\ s_{2,0} & s_{2,1} & s_{2,2} & s_{2,3} \\ s_{3,0} & s_{3,1} & s_{3,2} & s_{3,3} \end{bmatrix} = \begin{bmatrix} s'_{0,0} & s'_{0,1} & s'_{0,2} & s'_{0,3} \\ s'_{1,0} & s'_{1,1} & s'_{1,2} & s'_{1,3} \\ s'_{2,0} & s'_{2,1} & s'_{2,2} & s'_{2,3} \\ s'_{3,0} & s'_{3,1} & s'_{3,2} & s'_{3,3} \end{bmatrix}$$

Constant Tables

L Table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0		00	19	01	32	02	1A	C6	4B	C7	1B	68	33	EE	DF	03
1	64	04	E0	0E	34	8D	81	EF	4C	71	08	C8	F8	69	1C	C1
2	7D	C2	1D	B5	F9	B9	27	6A	4D	E4	A6	72	9A	C9	09	78
3	65	2F	8A	05	21	0F	E1	24	12	F0	82	45	35	93	DA	8E
4	96	8F	DB	BD	36	D0	CE	94	13	5C	D2	F1	40	46	83	38
5	66	DD	FD	30	BF	06	8B	62	B3	25	E2	98	22	88	91	10
6	7E	6E	48	C3	A3	B6	1E	42	3A	6B	28	54	FA	85	3D	BA
7	2B	79	0A	15	9B	9F	5E	CA	4E	D4	AC	E5	F3	73	A7	57
8	AF	58	A8	50	F4	EA	D6	74	4F	AE	E9	D5	E7	E6	AD	E8
9	2C	D7	75	7A	EB	16	0B	F5	59	CB	5F	B0	9C	A9	51	A0
A	7F	0C	F6	6F	17	C4	49	EC	D8	43	1F	2D	A4	76	7B	B7
B	CC	BB	3E	5A	FB	60	B1	86	3B	52	A1	6C	AA	55	29	9D
C	97	B2	87	90	61	BE	DC	FC	BC	95	CF	CD	37	3F	5B	D1
D	53	39	84	3C	41	A2	6D	47	14	2A	9E	5D	56	F2	D3	AB
E	44	11	92	D9	23	20	2E	89	B4	7C	B8	26	77	99	E3	A5
F	67	4A	ED	DE	C5	31	FE	18	0D	63	8C	80	C0	F7	70	07

E Table

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	01	03	05	0F	11	33	55	FF	1A	2E	72	96	A1	F8	13	35
1	5F	E1	38	48	D8	73	95	A4	F7	02	06	0A	1E	22	66	AA
2	E5	34	5C	E4	37	59	EB	26	6A	BE	D9	70	90	AB	E6	31
3	53	F5	04	0C	14	3C	44	CC	4F	D1	68	B8	D3	6E	B2	CD
4	4C	D4	67	A9	E0	3B	4D	D7	62	A6	F1	08	18	28	78	88
5	83	9E	B9	D0	6B	BD	DC	7F	81	98	B3	CE	49	DB	76	9A
6	B5	C4	57	F9	10	30	50	F0	0B	1D	27	69	BB	D6	61	A3
7	FE	19	2B	7D	87	92	AD	EC	2F	71	93	AE	E9	20	60	A0
8	FB	16	3A	4E	D2	6D	B7	C2	5D	E7	32	56	FA	15	3F	41
9	C3	5E	E2	3D	47	C9	40	C0	5B	ED	2C	74	9C	BF	DA	75
A	9F	BA	D5	64	AC	EF	2A	7E	82	9D	BC	DF	7A	8E	89	80
B	9B	B6	C1	58	E8	23	65	AF	EA	25	6F	B1	C8	43	C5	54
C	FC	1F	21	63	A5	F4	07	09	1B	2D	77	99	B0	CB	46	CA
D	45	CF	4A	DE	79	8B	86	91	A8	E3	3E	42	C6	51	F3	0E
E	12	36	5A	EE	29	7B	8D	8C	8F	8A	85	94	A7	F2	0D	17
F	39	4B	DD	7C	84	97	A2	FD	1C	24	6C	B4	C7	52	F6	01

S6 R

00	12	1	10	15	9	2	6	8	0	13	3	4	14	7	5	11
01	10	15	4	2	7	12	9	5	6	1	13	14	0	11	3	8
10	9	14	15	5	2	8	12	3	7	0	4	10	1	13	11	6
11	4	3	2	12	9	5	15	10	11	14	1	7	6	0	8	13

8421
1011

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Mid Term Examinations (Fall 2022)

Course Code with Title	**MS-203: Numerical Analysis**	**Program**	Bachelor of Computer Science BSCS
Instructor	1. Mr. Noman Uddin Qureshi 2. Ms. Nida Jamil, 3. Mr. Muhammad Ali	**Semester**	1st
Start date & Time	**07-12-2022 9:30 A.M**	**End Time**	**11:00 A.M**
Maximum Marks	**30**		

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.
- Don't write on paper use backside of copy for rough work.
- Pencil work will not be considered for marking.

Q:1 **(5)**

A projectile is fired with an angle of elevation. Find elapsed time by Using Newton's Raphson Method, the equations of motion

$y = f(t) = 4800(1 - e^{-t/10}) - 320t$

use the initial guess $t_0 = 8$. **(up to 4th iteration)**

Q:2 **(5)**

To determine the drag coefficient c (by False position Method) needed for a parachutist of mass m = 68.1 kg to have a velocity of 40 m/s after free falling for time t = 10 s.

Note: The acceleration due to gravity is 9.81 m/s^2

$$f(c) = \frac{mg}{c}\left(1 - e^{-\left(\frac{c}{6.81}\right)10}\right) - 40$$

Since

$f(12) = 6.114$ $\quad$ $f(16) = -2.230$ **(up to 4th Iteration)**

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Q: 3 **(5)**

Consider the system of equations

$$83x + 11y - 4z = 95$$
$$7x + 52y + 13z = 104$$
$$3x + 8y + 29z = 11$$

Solve by Gausses seidel iteration. up to three decimal places **(3^{rd} iterations)**

Q: 4 **(5)**

Find the real root of equation $x^3 - 9x + 1 = 0$ up to **three decimal places** by Fixed point Iteration method.

Q: 5 **(5)**

Solve Crout's Method

$$3x + y + z = 4$$
$$x + 2y + 2z = 3$$
$$2x + y + 3z = 4$$

Q: -6 **(5)**

Read following data carefully and find the value at x = 1895 by Newton's Forward difference formula

x	f(x)
1891	46
1901	66
1911	81
1921	93
1931	101

$y_0 + p\Delta y_0 +$

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Mid Term Examinations (Fall 2022)

Course Code with Title	**CS-325T: Artificial Intelligence**		**Program**	Bachelor of Computer Science BS(CS)
Instructor	**Ms. Rabiya Tahir/ Ms. Quratulain Khalid**		**Semester**	6^{th}
Start date & Time	December 05, 2022 at 09:30 AM	**End Time**	11:00 AM	
Maximum Marks	30 ABDUl REHMAN.			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions.

Q.1. **(1+1+1+1=04)**

Following are some application areas of artificial intelligence determine that they belong to which type of AI?

(a) Smart filter to identify and remove spam messages. Limited memory.

(b) Chatbots to impersonate the conversational styles of customer representatives.

(c) IBM's chess-playing supercomputer. Reactive

(d) Google's recommendation engine.

P - To Env

Q.2. **(6+1+1+2=10)**

The Wumpus is a computer game in which agent explores a cave consisting of rooms connected by passageways as shown in Fig: 01. Looking somewhere in the cave is the **Wumpus,** a **Beast** that eats an agent that enters its room. Some rooms contain bottomless pits that trap an agent that wanders into the room. Occasionally, there is a heap of gold in a room. The goal is to collect the gold and exit the world without being eaten. The agent always starts in the field [1,1]. The task of the agent is to find the gold, return to the field [1,1] and climb out of the cave. The agent perceives a stench in the squares adjacent to the sqaure containing wumpus, a breeze in the squares adjacent to the square containing pit, a glitter in the square where the gold, a bump if it walks into a wall and a woeful scream everywhere in the cave, if the wumpus is killed. The percepts will be given as a five-symbol list: If there is a stench, and a breeze, but no glitter, no bump, and no scream, the percept is **[Stench, Breeze, None, None, None].** The agent action is to go forward, turn right, turn left, grab means pick up an object that is in the same square as the agent, shoot means fire an arrow in a straight line in the direction the agent is looking. The arrow continues until it either hits and kills the wumpus or hits the wall, climb is used to leave the cave and die if the agent enters a square with a pit or a live wumpus.

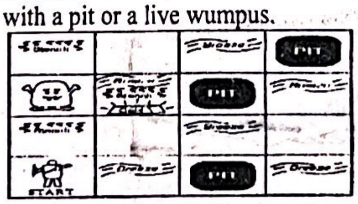

Fig: 01

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(a) Characterize the above scenario in terms of task environment by giving justification whether it is:

(i) Fully observable or partially observable.	(iv) Deterministic or nondeterministic.
(ii) Discrete or continuous.	(v) Static or dynamic.
(iii) Single-agent or multi-agent.	(vi) Episodic or sequential.

(b) Determine an agent percept at any given instant if it perceives no stench, breeze, glitter, no bump and no scream.

(c) Propose agent architecture among the following.

(i) Simple Reflex	(ii) Goal-Based
(iii) Utility-Based	(iv) Model Based

(d) Determine its Performance Measure, Environment, Actuator, Sensor.

Q.3. **(2+4+2=08)**

The graph shown in Fig: 02 starts from node "S" the goal node "G" will be achieved by exploring the number of nodes.

(a) Determine the search path by using Greedy BFS.
(b) Determine the search path by using A*.
(c) Which one is optimal in terms of path cost.

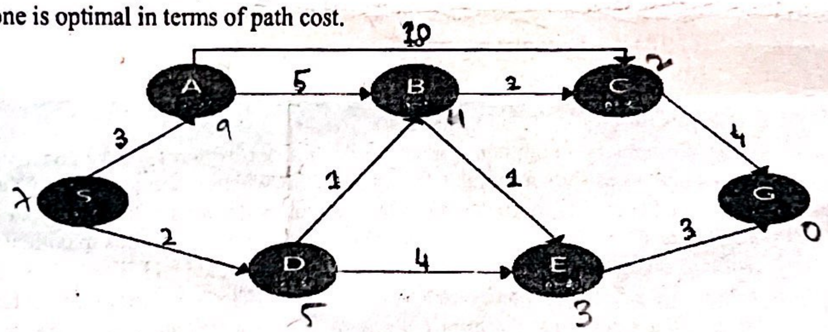

Fig: 02

Q.4. **(4+4=08)**

(a) Apply minimax algorithm on the given tree as shown in Fig: 03.
(b) Apply alpha beta pruning on the solution obtained in Part (a) to show which nodes will be pruned.

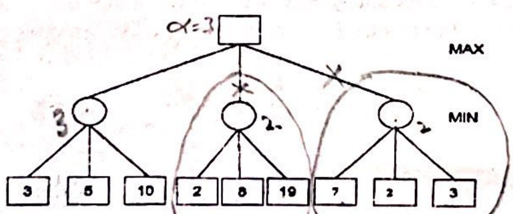

Fig: 03

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Mid Term Examinations (Fall 2022)

Course Code with Title	**CS-425: Design and Analysis of Algorithm**	**Program**	Bachelor of Computer Science BS(CS)
Instructor	**Ms. Ayesha Khaliq**	**Semester**	6^{th}
Start date & Time	6,December 2022, 9:30 AM	**End Time**	11:00 A.M
Maximum Marks	30		

IMPORTANT INSTRUCTIONS:
Read the following Instructions carefully:

- Attempt All Questions.

Q.1. **(8+2)**

a) Find the order of growth and show the upper and lower bound of given expressions

1. $10n^2 + 4n + 2$
2. $2n^2 + n \log n + 1$
3. $n\log(n^2+1) + n^2 \log n$
4. $T(n)=6*2^n + n^2$

b) Write an expression of following algorithm.

Algorithm 1	**Algorithm 2**
Function1 (n)	Function2 (a, n)
{ if n <= 1	{
then	sum=2;
output 'n'	a=4;
else	for i=1 to n
f2 = 0;	for j=1 to n-1
f1 = 1;	{
for i = 2 to n	sum=i*(sum+a[j]);
do	}
{ f = f1 + f2;	return sum;
f2 = f1;	
f1 = f; } output 'f' }	}

n + n(4n)+1
n + 4n²+1
2 + n(n+1)
2 + n² + n + 1
3 + n² + n.

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Q.2. **(6)**

A developer wants to calculate the computational time comparison between vectorized and non-vectorized method of two arrays one has element range with 10^5 and other contains the element range of 2×10^5. Write the code in python mentioning the comparison of time length using dot function of given arrays.

Q.3. **(4)**

How numerical python will be useful in the analysis of data algorithm. Justify your answer with the help of example.

Q.4. **(10)**

Solve the following recursive case till the base case is achieved

1

$$T(n) = \begin{cases} 0, & n = 1 \\ 0, & n = 2 \\ T(n-1) + T(n-2) + 1, & n \geq 3 \end{cases}$$

2

$$T(n) = \begin{cases} 1, & n = 0 \\ 3T(n-1), & n > 0 \end{cases}$$

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Mid Term Examinations (Fall 2022)

Course Code with Title	**CS-323T: Computer Graphics And Multimedia Applications**		**Program**	Bachelor of Computer Science BS(CS)
Instructor	**Mr.Faraz Khan and Mr.Faheem**		**Semester**	6^{th}
Start date & Time	December 09, 2022 At 09:15 AM	**End Time**	10:45 AM	
Maximum Marks	30			

IMPORTANT INSTRUCTIONS:

Read the following Instructions carefully:

- Attempt All Questions. Each question carries equal marks.
- Do Neat and clean work

Q.1. **(5+5)**

(a) Computer graphics and multimedia applications plays an important role in almost every professional field, list down any ten real life applications of multimedia and their importance in our daily life.

(b) Calculate coordinates of rectangle after **shearing** when shearing factors sx = 3 and sy = 3. Consider the vertices of rectangle as A(0,0), B(4,0),C(4,4) and D(0,4). Also draw the graph before shearing and after shearing.

Q.2. **(5+5)**

(a) How image plays an important role in multimedia applications. Differentiate the concept of bitmap image and vector image in detail, draw diagram of bitmap/raster image. Also discuss the desirable features of final year project poster (48x36 inch).

(b) Consider a triangle having vertices A(2,2), B(4,2) and C(3,4). Find the coordinates of triangle after a 90 degree clockwise and anticlockwise **rotation** around the origin. Also draw the graph.

Q.3. **(5+5)**

(a) A 640 x 480 pixel image of 16 bit colour would require how much disk space, discuss bitmap Fonts, also highlight concept of Tracking, Kerning, leading in text. Differentiate among linear text, non-linear text, rich text, hyper text, temporal media and non-temporal media.

(b) A circle having xc = 2, yc=5 with radius r = 10, calculate points/pixel of and show the steps in tabular form.